Lab 1 Exercise Solutions

SOCIOL 723, Social Statistics II

Wenhao Jiang

```r
library(dplyr)
gss <- readRDS("../Data/gss_earnings.rds")
fit <- lm(lnearn ~ educ + exper + I(exper^2) + female + black, data = gss)
```

Exercise 1: The dummy variable trap

```r
length(coef(lm(lnearn ~ factor(region), data = gss)))   # intercept + 8 dummies
#> [1] 9

## force all nine indicators in alongside an intercept
X9 <- model.matrix(~ 0 + factor(region), data = gss)
colnames(X9) <- paste0("r", 1:9)
coef(lm(gss$lnearn ~ X9))
#> (Intercept)        X9r1        X9r2        X9r3        X9r4        X9r5
#>    10.03949     0.09246     0.03436    -0.03353    -0.20571    -0.10960
#>        X9r6        X9r7        X9r8        X9r9
#>    -0.27817    -0.19337    -0.21375          NA
```

`lm()` fits nine coefficients, not ten: with `factor(region)` it drops one category automatically, and when we force all nine indicators in, it returns `NA` for the last one. The nine dummies sum to one for every respondent — exactly the intercept column — so once eight dummies and the intercept are in, the ninth is a perfect linear combination of them and carries no new information. `R` resolves the trap by refusing to estimate the redundant coefficient; the dropped category becomes the reference against which the others are measured.

Exercise 2: The $n - 2$, by simulation

```r
set.seed(1)
n_sim <- 2000
xs <- sample(8:20, n_sim, replace = TRUE)
```

```r
rss <- replicate(2000, {
  y <- 8.2 + 0.12 * xs + rnorm(n_sim, 0, 2)     # sigma^2 = 4
  sum(residuals(lm(y ~ xs))^2)
})
c(mean_rss = mean(rss),
  sigma2_x_nminus2 = 4 * (n_sim - 2),
  wrong_guess = 4 * n_sim)
#>         mean_rss sigma2_x_nminus2      wrong_guess
#>             8001             7992             8000
```

The average of $\sum_i \hat{e}_i^2$ sits on top of $\sigma^2(n-2)$, not $\sigma^2 n$. Fitting the two coefficients absorbs exactly two degrees of freedom's worth of error variation — the residuals are systematically a little smaller than the errors they estimate — and dividing by $n-2$ instead of n is precisely what makes s^2 unbiased.

Exercise 3: Sign the bias

Prediction first. The bias from omitting `hours` is $\beta_{\texttt{hours}} \times \delta_{\texttt{educ}}$, where $\delta_{\texttt{educ}}$ is the `educ` coefficient from the auxiliary regression of `hours` on the included regressors. More hours should mean higher earnings ($\beta_{\texttt{hours}} > 0$); whether the more educated work more or fewer hours is the genuinely uncertain sign.

```r
fit_h <- update(fit, . ~ . + hours)
aux   <- lm(hours ~ educ + exper + I(exper^2) + female + black, data = gss)

c(beta_hours = coef(fit_h)["hours"],
  delta_educ = coef(aux)["educ"])
#> beta_hours.hours  delta_educ.educ
#>         0.008609         0.032335
c(short_minus_long = coef(fit)["educ"] - coef(fit_h)["educ"],
  product          = coef(fit_h)["hours"] * coef(aux)["educ"])
#> short_minus_long.educ         product.hours
#>             0.0002784             0.0002784
```

Both pieces come out positive — conditional on the controls, the more educated work slightly more hours, and hours pay — so the short regression *overstates* the schooling coefficient. The OVB identity holds to machine precision, but note the magnitude: the bias is well under one percent of the coefficient. Signing a bias and sizing it are different exercises.

Exercise 4: The FWL standard error

```r
d_tilde <- residuals(lm(educ ~ exper + I(exper^2) + female + black, data = gss))
s2 <- sum(residuals(fit)^2) / (nrow(gss) - length(coef(fit)))
```

```r
c(fwl     = sqrt(s2 / sum(d_tilde^2)),
  summary = coef(summary(fit))["educ", "Std. Error"])
#>      fwl  summary
#> 0.005103 0.005103
```

Identical. The standard error on `educ` in the full regression runs on the *residualized* variation $\sum_i \tilde{d}_i^2$ — the part of schooling the controls cannot predict — which is why adding controls correlated with `educ` inflates its standard error even as it sharpens its interpretation.

Exercise 5: Saturate the key covariate

```r
fit_sat <- lm(lnearn ~ factor(educ) + exper + I(exper^2) + female + black,
              data = gss)
anova(fit, fit_sat)
#> Analysis of Variance Table
#>
#> Model 1: lnearn ~ educ + exper + I(exper^2) + female + black
#> Model 2: lnearn ~ factor(educ) + exper + I(exper^2) + female + black
#>   Res.Df  RSS Df Sum of Sq    F           Pr(>F)
#> 1   3503 2350
#> 2   3485 2286 18      64.4 5.46 0.00000000000074 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The model comparison rejects decisively: freeing every year of schooling to have its own coefficient buys more fit than its extra parameters cost, so the earnings–schooling relationship is not exactly linear (the jumps at 12 and 16 years — credential effects — are the usual culprits). That is not a verdict against the linear model: the single slope remains an excellent one-number summary, and Week 1's lesson applies — a line fit to a curved relationship estimates the best linear approximation to it. Saturating is how you check what that approximation is smoothing over.